Exact stopping distance six in a QC-GHP cyclic-cover family

24 October 2026 · Verified checkpoint · Follow-up to L-F16-stopping-bound

Abstract. For every positive integer t, we give explicit binary CSS check matrices of length 882t whose X-check graph has stopping distance exactly six. The matrices use lift length 63t and the fixed polynomials displayed below. A sound degree-one extension search, reduced to two roots and evaluated in Lean's kernel, proves the base lower bound; a checked local-cover argument transfers it to every member of the family. A companion Z-check row gives the upper witness. We also prove that the actual expanded X-matrix has second squared singular value at least 12. Consequently the source spectral condition fails, and both Zhu–Chugg two-eigenvalue expressions are negative whenever their strict gap holds. All new proofs and source-to-matrix bridges have checked Lean versions.

1. Question, source and precise answer

Abiad and Castriota [1], 15 July 2026, arXiv:2607.13666v1, Proposition 5.1, Remark 5.2 and Section 6, ask for QC-GHP parameters making their spectral stopping bound informative and for tighter bounds. The preceding research block establishes a scalar-factor obstruction to the first route. This paper answers our narrower, incidence-aware follow-up: can actual incidences certify the exact stopping distance of the running B1 matrix and its positive 63-multiple covers?

Theorem 1. Let t be a positive integer, $\ell=63t$, and let P(s) have entries $P(s)_{i,j}=1$ exactly when $i=j+s$ in Z/ℓ. Let S be the forward cyclic shift of order 7. Over $F_2[x]/(x^{\ell}-1)$, put

$$A=x^{27}I_7+S+x^{54}S^2, \quad b=1+x+x^6,$$
$$H_X=[\tilde{A} \mid I_7\otimes\tilde{B}^T], \quad H_Z=[I_7\otimes\tilde{B} \mid \tilde{A}^T].$$

Then $H_XH_Z^T=0$ over F_2. Each matrix has 882t columns and 441t rows; H_X has column weight 3 and row weight 6. Every nonempty H_X stopping support has size at least six, and an H_Z row is a size-six stopping support. Thus the H_X stopping distance is six. For the real expanded matrix, $\sigma_2(H_X)^2\geq 12$, counting Gram eigenvalues in descending order with multiplicity.

Scope. These are CSS check matrices. We do not establish logical dimension, logical distance, rate or decoding performance for the covers. A stabilizer support is not a logical operator. At t=1 the displayed matrices are the source's Example 3.2; its reported [[882,24,d]], $18\leq d\leq 24$ parameters are cited, not reproved here. The family does not satisfy Proposition 5.1's spectral inequality.

2. A sound lower certificate on the base

For a binary incidence relation E(v,c), a stopping support is a nonempty finite variable set T such that no check has exactly one neighbor in T. No elementary, connected-support, or even-syndrome hypothesis is imposed. With checks (j,f) in $Z/7\times Z/\ell$ and variables (side,j,f), the source matrices give precisely these column lists:

left(j,f): (j,f+27), (j+1,f), (j+2,f+54);
right(j,f): (j,f), (j,f−1), (j,f−6).

Lemma 2 (extension soundness). Suppose S is a duplicate-free list contained in a stopping support T. If a check has one neighbor in S, it has a second neighbor in T outside S. Branching over every other neighbor of that check therefore preserves at least one branch contained in T. If $|T|\le|S|+f$, f further additions suffice to reach a partial set with no degree-one check. Consequently rejection of every branch excludes all such stopping supersets.

Proof. If no check has degree one in S, the test accepts. Otherwise the stopping property supplies the extra neighbor. Append that neighbor, retain duplicate-freeness and containment, reduce the fuel by one, and apply induction. Completeness of the explicit row list and the check picker's degree-one condition are proved separately. An accepting branch is not used to infer a lower bound. □

Lemma 3 (two roots). Simultaneously translating both cyclic coordinates of all variables and checks preserves incidence, support cardinality, and the stopping property. Every selected variable translates to (side,0,0). Thus it suffices to exclude small stopping supports containing either of the two side representatives.

Proof. Each entry in the displayed incidence lists is affine with coefficient one in the translated coordinates. Translation is a bijection on variables and on checks. Image cardinalities and the selected-neighbor cardinality at every check are therefore preserved. Nonemptiness supplies the selected variable. □

Base certificate. At $\ell=63$ the checked Boolean extension test rejects both roots with four further additions. Lemmas 2–3 exclude every stopping support of size at most five. This is an exact finite certificate, evaluated by **decide +kernel**; neither external enumeration nor native evaluation is a proof premise.

The independent Python reconstruction also searched all 882 base roots, using 688,842 search nodes, without finding a support of size at most five. This corroborates the model and implementation. The Lean certificate and its soundness theorem supply the proof.

```
normalized_roots_reject :
∀ b : Bool, survivesList pickList row 4 [(b,0,0)] = false


running_lower_six (T : Finset Var) (hT : Stopping E T) : 6 ≤ T.card
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The search strategy is classical rooted narrowing and branch-and-bound; Wang–Kulkarni–Poor [3] are relevant prior art. The contribution here is the checked instance, exact source translation and cover extension, rather than a new general enumeration algorithm.

3. Cover transfer, upper witness and CSS bridge

Lemma 4 (local-cover transfer). Let variable and check maps take a graph to a base graph. Assume: incidences map to incidences; every base check incident to the image of a selected variable has an incident lift at that variable; and the variable map is injective on each check's neighbor set. Then the image of every stopping support is a stopping support, with cardinality at most the original support.

Proof. The image is nonempty. If a base check meets it, choose a selected variable over that meeting and lift the check. The original stopping property provides a different selected neighbor at the lifted check. Local injectivity makes their images distinct; incidence preservation puts both at the base check. Hence the base check cannot have selected degree one. The finite-image cardinality inequality gives the size claim. □

For $\ell=63t$, reduction $\mathbb{Z}/\ell\to\mathbb{Z}/63$ maps both explicit column lists onto their base lists. The six neighbors of any lifted check map to six distinct base neighbors. The reduction homomorphism, incidence preservation, lifting and local injectivity are all proved in Lean. Lemma 4 and the base lower certificate give the lower bound six for every positive t; this is not extrapolation from sampled lift lengths.

Upper witness. For every such lift the following support has six distinct variables:

left(0,0), left(0,−1), left(0,−6),
right(0,27), right(1,0), right(2,54).

Every incidence of a selected variable has an explicitly checked selected partner on the other side, so no check has selected degree one. The support is exactly H_Z row (0,0). For the base, kernel reduction checks that every induced check degree is zero or two. These facts give the upper bound six. Combining it with the lower bound proves Theorem 1's exact stopping distance.

Actual matrix bridge. The formalization first defines the displayed matrices over the naturals. Duplicate-free column lists imply their entries are zero or one, and their count predicates are proved equivalent to the incidence lists. Casting these entries to F_2 preserves precisely the same graph. Separate counting lemmas prove the dimensions and H_X degrees. Thus the stopping theorem concerns the actual binary matrices, not only an abstract graph model.

CSS proof. On the finite abelian product $\mathbb{Z}/7\times\mathbb{Z}/\ell$, $\tilde{A}$ and $D=I_7\otimes\tilde{B}$ are circulant matrices. Circulant commutation gives $\tilde{A}D^T=D^T\tilde{A}$. The expanded binary product is $\tilde{A}D^T+D^T\tilde{A}$, which vanishes in characteristic two. All identities, transpose conventions and casts are checked for the displayed matrices.

The existence of stabilizer stopping supports is known in quantum erasure decoding [4,5], with the mechanism attributed there to Connolly et al. [6]. The witness is an instance of that mechanism. The exact lower bound and the cover comparison are separate claims; the witness alone does not establish them.

4. Spectral comparison and known stopping sets

Theorem 5. For every positive lift in Theorem 1, the actual real matrix has $\sigma_2^2 \geq 12$.

Proof. Consider the two-dimensional plane of row vectors constant on fibers, supported at block rows 0 and 3, with real coefficients a and b. Its squared norm is $\ell(a^2+b^2)$. Direct sums over the expanded matrix show that the left block contributes $3\ell(a^2+b^2)$ and the right block contributes $9\ell(a^2+b^2)$. The energy is therefore 12 times the squared norm. The linear functional giving the top eigenvector coefficient has a nonzero kernel on this plane. A nonzero vector in that kernel has Rayleigh quotient at most the second Gram eigenvalue, including multiplicity. Hence that eigenvalue is at least 12. All finite sums and the spectral reduction are checked in Lean. □

Proposition 5.1 of [1] requires $\sigma_2^2 < \gamma/2$; here $\gamma=3$ and $12>3/2$, so the condition fails. In the preceding block, inspection of the exact radicals and the separate exclusion of the small-size branch also showed why satisfying the spectral inequality alone need not give a useful stopping-distance guarantee. We do not apply that proposition outside its hypotheses.

For comparison, Zhu–Chugg [2], Corollary 6, equations (41)–(42), give the following two regular-graph stopping bounds, with N variables, column degree γ, row degree k and second Gram eigenvalue η:

$$N(2\gamma-\eta)/(\gamma k-\eta),$$
$$2N(2\gamma+k-2-\eta)/(k(\gamma k-\eta)).$$

Their strict spectral-gap regime is $\eta<\gamma k$, with the connected regular setting supplying the unique top eigenvalue. These formulas have no separate small-support branch. They use a different threshold from [1]. In our family $N=14\ell$, $\gamma=3$, $k=6$ and $\eta\geq 12$. If $\eta<18$, both denominators are positive but the numerators have factors $6-\eta$ and $10-\eta$, both negative. Thus both expressions are negative. If the strict gap fails, the displayed bounds are outside that regime. The two cases require no numerically inferred connectivity or spectral gap. This comparison does not dismiss Zhu–Chugg's broader multi-eigenvector or irregular analysis.

As an illustration only, independent dense reconstruction at $\ell=63$ gives $\eta\approx 15.89294145$, and the two formula values ≈ -4141.1162 and ≈ -822.2481. Floating-point values do not enter the proof.

Prabhu et al. [4] distinguish stabilizer, horizontal and vertical stopping supports and study erasure decoders. Their constituent expansion conditions are not spectral conditions on expanded H_X. Yao et al. [5] explicitly simulate B1 and study post-peeling cluster decoding. The inspected full texts do not supply the lower certificate proved here. Plain peeling obstructions and quantum-aware correctability are different questions.

The 2026 HGP stopping-set paper [7] remains a full-text access gap; its abstract is insufficient for a complete theorem comparison. The public code and X searches were bounded; they support no global absence or priority claim. The detailed search and source-screening ledgers accompany the proof bundle.

5. Formal evidence and reproducibility

The aggregate theorem **Incidence.Family.positive_family** quantifies over every natural t with [NeZero t] and combines dimensions, H_X degrees, actual binary CSS commutation, the universal lower bound for stopping supports and an existential size-six witness. The source B1 consumer specializes it to t=1. The spectral theorem uses the descending ordered Gram spectrum of the same expanded real matrix.

Claim / proof	Lean source and main declaration
Stopping definition; extension; translation	IncidenceCore.lean: stopping_extension, shift_incidence, stopping_image
List-search soundness and picker	IncidenceList.lean: survivesList_of_stopping, pickList_singleton
Kernel certificate and base lower bound	Incidence.lean: normalized_roots_reject, running_lower_six
Abstract cover theorem	IncidenceCover.lean: stopping_cover_image
Concrete cover; six-element support	IncidenceFamilyCore.lean: maps, lifts, local_inj, six_card, six_stopping
Base source matrices and profile	IncidenceSource.lean: HX_incidence, six_profile, six_is_HZ_row
Family source/binary and companion-row bridge	IncidenceFamilySource.lean: HX_binary, six_is_HZ_row
Cover lower bound / exact natural matrix	IncidenceFamily.lean: lower_six, source_matrix_exact_six
Actual binary CSS and incidence	IncidenceCSS.lean: css, HX2_incidence
Degrees and dimensions	IncidenceDegrees.lean: column_weight, row_weight, dimensions
Expanded spectrum and Zhu–Chugg comparison	IncidenceSpectral.lean: plane_energy, second_lower_twelve, zhu_chugg_bounds_negative
Combined positive family and source consumer	IncidenceCertificate.lean: positive_family, source_B1_stopping_distance_six

Pinned environment: Lean 4.36.0-rc1; mathlib commit 5f7b4198d22a9c82e1c698a379334fe128095c82. The standalone project vendors the parent Spectral.lean unchanged, SHA-256 a3e712b2a206bd7743408ecc122e66e7d980dbd4689c6d49977536dfe9bb469d. It is inherited infrastructure, not a newly counted result. The fresh aggregate build completed successfully (5,494 jobs). The accompanying audit checks every named declaration against the standard axiom allowlist. Exact commands, signatures, build log and hashes are in the archive.

Reproduce: **lake build IncidenceCertificate**; then **lake env lean AuditIncidence.lean**. README.txt explains cache setup. Python cross-checks reconstruct the integer matrix independently, exhaust all 882 base roots, and test lift lengths 63t for t=1,...,8. They corroborate the formal result; the universal theorem is established in Lean.

This checkpoint is local and uncommitted. Public commit-pinned links for the new files do not yet exist; the archive’s hashes are the reproducibility anchor. Planned folder: github.com/alectionary/vibe-math, formalization/stopping-bound/incidence/. Complete file links are in PROOF-LINKS.html inside the source archive. No publication or X posting has occurred.

6. Chronology, inventory and limitations

Source	Relevant date / inspected milestone
Tanner classical spectral ingredient	Inspected manuscript 26 July 2000; later 2001 publication not independently inspected
Zhu–Chugg	28 September 2005 conference metadata; December 2007 journal inspected (month precision)
Wang–Kulkarni–Poor	19 May 2006 preprint; inspected journal version 22 April 2009 / May 2009
Panteleev–Kalachev construction	4 April 2019 preprint; 12 November 2021 v3 inspected in parent block
Yao–Gokduman–Pfister	11 December 2024 v1 inspected
Prabhu et al.	30 April 2025 v1; 29 May 2025 v2 inspected
Abiad–Castriota question [1]	15 July 2026 v1
This checkpoint	24 October 2026; 101 calendar days after [1]

The 101-day difference is date arithmetic, not an assertion about how long a problem remained unsolved. The original investigation began at 19:49 UTC; the incidence follow-up began at 20:48 UTC on 24 October. Verification and manuscript checkpoint times are recorded separately in verification.json. Classical ingredients' ages do not measure the age of this exact question.

Three additional accessible full texts were screened for open questions. Five broad historical directions enter the shared inventory with current resolution and Lean status unassessed; one 2026 paper retains unknown question count due to missing full text. The cohort moves from 262 to 267 canonical entries, with 135 eligible targets and four completed canonical Lean targets unchanged. This restricted project-derived follow-up supports L-F16; it is not another completed author-posed target.

Question answered. The displayed B1 X-check matrix and all positive 63-multiple cyclic covers have stopping distance exactly six, with checked source, CSS, cover and spectral bridges. **Question still open.** A broadly applicable incidence-conditioned spectral/design bound for other QC-GHP parameters, the higher-rank construction, and the logical ranks/distances of these covers remain outside this result. **Next proposed target (follow-up).** Classify all size-six base supports and identify which are companion-row stabilizers; this bounded target reuses the sound search infrastructure. It has not been completed here.

References

[1] A. Abiad and N. Castriota. Spectral and Additive Combinatorial Methods for Cycles and Absorbing Sets in Lifted-Product Quantum LDPC Codes. arXiv:2607.13666v1 (15 July 2026), Example 3.2, Proposition 5.1, Remark 5.2, §6. https://arxiv.org/html/2607.13666v1

[2] M. Zhu and K. M. Chugg. Bounds on the Expansion Properties of Tanner Graphs. IEEE Transactions on Information Theory 53(12), 4834–4840 (December 2007), Corollary 6. https://hal.usc.edu/chugg/docs/pubs/zhu2007bounds.pdf

[3] C.-C. Wang, S. R. Kulkarni and H. V. Poor. Finding All Small Error-Prone Substructures in LDPC Codes. IEEE Transactions on Information Theory 55(5), 1976–1999 (May 2009). DOI:10.1109/TIT.2009.2015993.

[4] Prabhu et al. arXiv:2504.21845v2 (29 May 2025), §III.B–D and §VII. https://arxiv.org/abs/2504.21845v2

[5] Yao, Gokduman and Pfister. arXiv:2412.08817v1 (11 December 2024), §III and Figure 7. https://arxiv.org/abs/2412.08817v1

[6] Connolly et al. Quantum 8, 1450 (27 August 2024). DOI:10.22331/q-2024-08-27-1450. Metadata and cited stabilizer mechanism only; full text not independently screened in this follow-up.

[7] Prabhu, Gayathri and Kumar. On the Analysis of Stopping Sets Associated with the Quantum Hypergraph Product Code. ISIT 2026. DOI:10.1109/ISIT62367.2026.11654127. Named full-text access gap.